
UNIT 17 THEORY OF MONETARY OF POLICY*

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17.0 OBJECTIVES

After going through this unit you would be able to:

- explain the discretion and rule of the Central Bank in determining inflation and output;
- understand the current consensus of monetary policy;
- assess the impact of a higher inflation target on interest rate and aggregate demand;
- assess the impact of an expansionary fiscal policy on interest rate, inflation and aggregate demand;
- explain the path dependence in macroeconomic outcomes.

17.1 INTRODUCTION

In this unit we focus on three main themes related to the conduct of monetary policy. First, we discuss the rules vs. discretion debate. Discretionary policy is arbitrary and often mis-placed (policy makers may follow an expansionary policy while the need is for a contractionary monetary policy). Further, discretionary monetary policy results in various types of lags (such as identification lag, decision lag and implementation lag). Thus, policy rules are preferred to discretionary measures by mainstream economists. However, there appears to be no consensus on the definition of rules. According to Friedman, Central Banks

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should follow the rule of constant monetary growth over time. Other economists such as Robert Barro define the rules as a commitment to zero/low inflation policy. For most new-Keynesian economists, policy rules are shown in terms of equations that link the Central Bank's policy rate (i.e., the key lending rate of the Central Bank; repo rate, for example, in India) to deviations from inflation and output targets.

Secondly, we will discuss about the current consensus on monetary policy. In the consensus model, monetary policy plays a major role in fixing inflation rate at its pre-defined target. In other words, monetary policy can successfully provide the nominal anchor. Moreover, monetary policy, while successful in determining the rate of inflation, cannot influence the level of output, which is fixed at its natural (i.e., full employment) level. In this sense, the new consensus is simply a revised version of Friedman's idea that market forces by themselves deliver full employment and monetary policy is neutral. At best, Central Banks can hope to reduce fluctuations of output around its natural level by following an appropriate interest rate policy.

Finally, we will talk about the criticism of the new consensus model. The new consensus model assumes that natural output is fixed. Critics point out that this may not be so. High aggregate demand (injected through monetary or fiscal means) may increase actual as well as natural output levels and take the economy to a higher equilibrium. We will use the concepts of IS curve, expectations augmented Phillips Curve, rational expectations, and adaptive expectations. You are familiar with these concepts from the earlier Units.

17.2 RULES, DISCRETION AND DYNAMIC CONSISTENCY

As we saw in Unit 2, Keynes advocated active government intervention. When there is shortfall in effective demand, the state was expected to increase government expenditure, may be through a deficit budget. Similarly, when the economy is passing through inflation, the government should go for a surplus budget. Such variation in government expenditure necessitated certain discretionary policy measures by the government. The classical economists, on the other hand, had suggested minimum government intervention in the economy so that market forces (demand and supply) can operate efficiently. The new classical economists suggest certain policy rules on macroeconomic variables.

It may be useful to discuss Friedman's (1968) prescription of what monetary policy should seek to achieve. First, Friedman weighed in against the then popular belief in trade-off between inflation and unemployment. According to Friedman, no such trade-off exists in the long run and the attempts to push output above its 'natural' level will only yield higher rates of inflation. As we shall see in the next section, this part of Friedman's analysis plays a very important role even in the current consensus.

Secondly, Friedman believed that Central Banks do not use their discretion

entirely judiciously. For example, the Federal Reserve in the US contracted money supply during the Great Depression. When Central Banks did not err in the wrong direction, they did so by moving too far, thereby, causing wide swings in money supply. Thus, Friedman proposed a ‘publicly stated policy of steady rate of growth of money supply’ for Central Banks. Moreover, the constant money growth rule is superior to other rules that targeted a certain price level or inflation rate. This is because ‘we cannot predict accurately just what effect a particular monetary action has on the price level and, equally importantly, just when it will have that effect’. Most Central Banks and economists today do not share Friedman’s confidence in the ability of monetary authorities to fix money supply. They also do not share his pessimism regarding inflation targeting. Where the two – Friedman and the current mainstream – do converge is in their shared belief in the superiority of rules over discretion.

17.2.1 Loss Function

There is a belief that while rules are superior to discretion, they may be dynamically inconsistent – having announced the rule, Central Banks may subsequently find it optimal to violate it. Barro and Gordon’s work is an important contribution in this area. Let us look at their main contributions through a simple model. First, let us introduce the loss function of a Central Bank.

The main objective of monetary policy is inflation targeting but control of inflation should not be at the cost of economic growth and employment. We assume that the economic welfare of the country is the maximum when the economy is operating at its natural output level (i.e., full employment output level) and inflation is at the targeted rate. There is a loss of welfare if inflation is different from the targeted rate and actual output is different from the natural output level. If we consider both these issues, we obtain a *loss function* of the Central Bank, given by

$$L = x^2 + (y - y'')^2 \quad \dots (17.1)$$

In equation (17.1), we denote inflation rate by x and output by y . The notation y'' is the output target pursued by the Central Bank. Let us denote natural output level by y' . Natural output stands for the output level that will be produced when labour force is fully employed. Let us assume that the Central Bank sets its output target in excess of the natural output level, y' . The Central Bank dislikes any inflation and deviations of output from its target. Higher value of the loss function (L) implies greater loss or welfare cost suffered by the economy.

The Phillips Curve is given by the following equation:

$$x = x^e + e(y - y') \quad \dots (17.2)$$

Equation (17.2) says that actual inflation (x) will exceed expected inflation (x^e) whenever actual output (y) exceeds natural output (y'). Objective of the Central Bank is to choose that combination of output and inflation that minimises its loss function. Obviously, any such combination must lie on the Phillips Curve. Thus,

the Central Bank minimises its loss function subject to the constraint of equation (17.2). The Central Bank tries to reach its optimal solution by varying either the money supply or the rate of interest. We can add another equation to the model that relates aggregate demand (which equals output y) to money supply and inflation rate. The optimal combination of output and inflation can then be inserted in the aggregate demand equation to obtain optimal money supply. Without intending to carry out such an exercise, we simply assume that the Central Bank moves some monetary policy instrument under its control to reach its desired outcome.

The Central Bank's optimisation exercise is similar to that of a household in consumer theory. Just as the consumer tries to reach the highest possible indifference curve for a given budget constraint, the Central Bank tries to attain the lowest possible loss function for a given Phillips Curve. Fig. 17.1 shows the relevant picture. Loss functions of the Central Bank are given by concentric circles centred at B, where inflation rate is zero and output is at its target y'' . The point B in Fig. 17.1 stands for the 'bliss point' as the Central Bank can do no better than to achieve its output target at zero inflation. The larger is the circle, the greater is the welfare cost incurred by the Central Bank. At each point on the vertical axis, output is at its natural level and, from the Phillips Curve equation, actual and expected inflation rates coincide. Since actual output increases with the inflation rate, the Phillips Curve is an upward-sloping line.

At an optimum point, the slope of the loss function is equal that of the Phillips Curve. Let the slope of the Phillips Curve be equal to e . The slope of the loss function is given by $e = \frac{(y'' - y)}{x}$ (you can verify this by totally differentiating equation (17.1) and setting $dL = 0$). Hence, at the optimum point:

$$ex = y'' - y \quad \dots (17.3)$$

After multiplying both sides of equation (17.3) by e and adding with equation (17.2), we obtain:

$$x = [x^e + e(y'' - y')]/(1 + e^2) \quad \dots (17.4)$$

In other words, the Central Bank's choice of inflation rate will vary positively with its output target, y'' and with the inflation rate expected by public, x^e .

17.2.2 Rules vs. Discretion

We now endogenise expected inflation rate by assuming that public's expectations are rational. That is, the public is aware that the Central Bank minimises its loss function subject to the constraint posed by the Phillips Curve and, thus, fixes inflation rate according to equation (17.4). Therefore, public will come to expect the inflation rate that the Central Bank actually fixes. This implies the following equation:

$$x = x^e = [x^e + e(y'' - y')]/(1 + e^2) \quad \dots (17.5)$$

On solving the above equation, we obtain:

$$x = x^e = (y'' - y')/e \quad \dots (17.6)$$

Equation (17.6) gives the discretionary outcome. Inflation rate under discretion is endogenous. It depends on the output target set by the Central Bank. A higher output target of the Central Bank results in a higher rate of inflation but fails to stimulate output above its natural level. This is because output can be stepped above its natural level only by increasing the inflation rate above its expected level. But the Central Bank cannot create any inflation surprise as long as the public forms its expectations rationally. When the Central Bank uses its discretion to step output above its natural level, all it manages to achieve is a positive inflation rate. In Fig. 17.1, the discretionary outcome is depicted at A. At this point two conditions are fulfilled: (i) slope of the loss function equals the slope of the Philips Curve, and (ii) actual inflation rate equals the expected inflation rate.

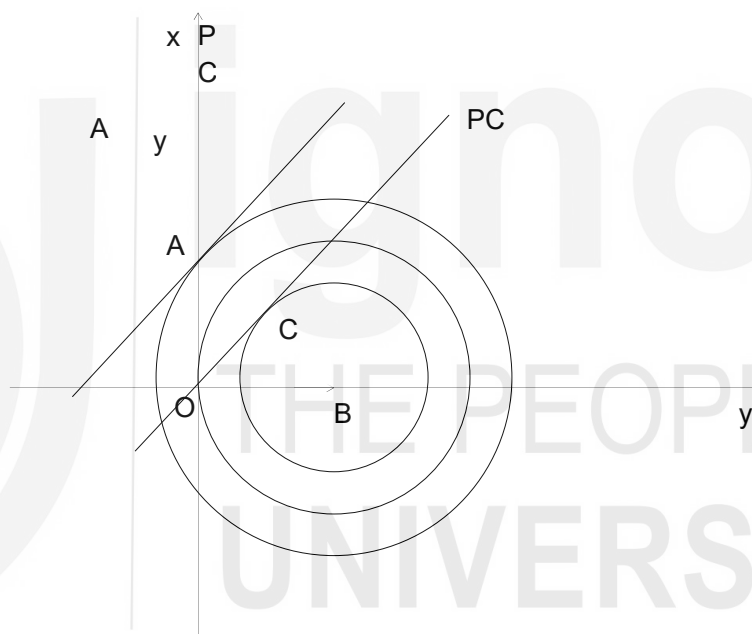


Fig. 17.1: Rules vs. Discretion

Now, suppose the Central Bank shifts from a discretionary policy framework to rule-based policy framework. Instead of arriving at certain positive inflation rate after solving its minimisation exercise, suppose the Central Bank declares a 'zero inflation rule'. Given rational expectations, the public will expect the rule that the Central Bank intends to increase production. Therefore, under the zero inflation rule:

$$x_r = x^e = 0 \quad \dots (17.7)$$

where the subscript r stands for 'rule'.

17.2.3 Dynamic Inconsistency

Given our formulation of the Phillips Curve, equation (17.5) implies equality

between actual and natural output. The rule-based solution produces the same level of output as the discretionary solution but at a lower inflation rate. Therefore, the ‘zero inflation rule’ implies a lower welfare cost than the discretionary solution. The rule-based solution is depicted by point O in Fig. 17.1. Now, suppose the Central Bank anchors expected inflation rate at zero by declaring a ‘zero inflation rule’. Since actual inflation equals expected inflation on the vertical axis, the relevant Phillips Curve (PC) passes through O in Fig. 17.1. It is clear that the Central Bank can do better than point O if it *cheats* on its commitment by choosing point C on the PC line where its welfare cost is minimised. Since the Central Bank’s minimisation exercise yields equation (17.5), we can find the relevant inflation rate by inserting $x^e = 0$ in equation (17.5). This gives:

$$x_c = e(y'' - y')/(1 + e^2) \quad \dots (17.8)$$

where c stands for cheating.

Since $y'' > y'$, equation (17.8) implies a positive inflation rate. The intuition is simple. The Central Bank first anchors the public’s inflationary expectations by announcing a ‘zero inflation rate’ policy and then creates an inflation surprise by violating its rule. Given our formulation of the Phillips’ Curve, this causes the output level to rise above its natural level. Also, the Central Bank has convex preferences; it prefers averages to extremes. Therefore, the combination of zero output gap (excess of actual over natural output) and zero inflation rate entails a greater welfare cost than a point like C where both inflation rate and output gap are positive.

The discussion so far can be summed up as follows: Rules, though superior to discretion, suffer from dynamic inconsistency. Having announced a rule, the Central Bank would be tempted to violate it. The problem is that once the public’s trust is broken, the public would come to believe that the Central Bank, regardless of what it proclaims, would apply its discretion in fixing the inflation rate. The Central Bank would then maximise its welfare by taking public’s inflationary expectations as given. The Central bank may discard its rule-based policy and end up with the discretionary case. Such shift in policy, however, should be avoided; because such change would entail inflation with no gain in output.

Mainstream economists maintain that delegation of monetary policy-making to conservative Central Bankers, who place a high weight on inflation control, can help to address the dynamic inconsistency problem. The benefits of delegation to the conservative Central Banker were captured in a model developed by Kenneth Rogoff. Suppose the conduct of monetary policy is delegated to an individual whose loss function is given by:

$$L = \gamma x^2 + (y - y'')^2 \quad \dots (17.9)$$

where $\gamma > 1$.

The new loss function places a higher weight on inflation. The conservative

Central Banker minimises the objective function given by equation (17.9) subject to the constraint given by the Phillips Curve (see, equation (17.2)). The public believes in rational expectations, and thus, expected and actual inflation rates are equal. The solution under discretion is now given by:

$$x = \frac{y'' - y'}{\gamma e} \quad \dots (17.10)$$

The intermediate steps involved in derivation of equation (17.10) are skipped here. We leave it as an exercise for the reader (hint: repeat the steps that were earlier followed in deriving equation (17.6)). Since $\gamma > 1$, inflation rate falls after monetary policy is delegated to a conservative Central Banker (this can be verified by comparing equation (17.10) with equation (17.6)). The government, which delegates policy-making to a conservative Central Banker will benefit, since a lower rate of inflation also reduces the value of the government's loss function given by equation (17.1).

Of course, delegation of policy-making to a conservative Central Banker will work only when the banker is independent of government control. The government may not be as inflation-averse as the Central Banker (if it is as inflation averse, nothing is gained from delegation). Thus, delegation alone will not lower the inflation rate; the conservative Central Banker must be free from the control of the government to pursue the objective of inflation control.

Check Your Progress 1

1. Explain why policy rule is preferred to discretion.

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2. What is meant by dynamic consistency?

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17.3. CONSENSUS VIEW OF MONETARY POLICY

We start the analysis by describing the main objectives of a Central Bank. We hypothesize that the Central Bank has an inflation target of x' in mind. It also wishes to keep output at its natural level, y' . The main instrument of the Central Bank policy is the rate of interest. In older versions of many textbooks, it is

assumed that the Central Bank pursues its objectives by regulating the supply of money. Under this policy, the interaction between supply of money and demand for money determines the rate of interest. An implication of the above is that money supply is an exogenous variable (determined by the Central Bank) while interest rate is an endogenous variable.

In recent years, the consensus has shifted in the other direction. Most economists today argue that the main instrument of monetary policy is interest rate. For example, the Reserve Bank of India fixes its repo rate (the rate at which it lends to commercial banks), which in turn influences both loan and deposit rates in the economy. Thus, we assume that the interest rate is exogenous and money supply is endogenous.

The main variable that influences aggregate demand is the real rate of interest, i.e., the excess of nominal interest rate over (expected) inflation. We shall assume that the Central Bank can influence the short term real interest rate by varying the short term nominal interest rate. A question arises: how does the Central Bank decide on the interest rate?

In macroeconomic literature it is well established that the Central Bank fixes the (real) rate of interest according to the following formula:

$$r = r' + a(x - x') + b(y - y') \quad \dots (17.11)$$

The formula given at equation (17.11) is called the Taylor's rule. Equation (17.11) assumes that both a and b are positive. It implies that the rate of interest is positively related to deviation of output ($y - y'$) and deviation of inflation ($x - x'$) from their target levels. When output and inflation are at their target levels, the rate of interest will be equal to r' . If actual inflation is higher than targeted inflation (or, actual output is higher than natural output), the Central Bank will increase the rate of interest. You would have noticed this in the decisions of the Monetary Policy Committee of the RBI.

The demand side of the economy is represented by an IS curve of the following type:

$$y - y' = c - dr \quad \dots (17.12)$$

Equation (17.12) indicates that actual output and, thus, the output gap responds negatively to real interest rate (r), where c captures the effect of other factors such as fiscal policy and external demand on output. Notice that the output gap will be zero when $r = \frac{c}{d}$. The rate of interest (r), at which output gap equals zero is called as the 'neutral rate of interest'. Both c and d are, of course, positive.

Inflation and output are related to each other through an "accelerationist" Phillip's Curve. Inflation is of the demand-pull variety. Inflation increases over time when high aggregate demand causes output to remain above its natural level (this requires e to be positive in equation (17.13)). This is captured by the following simple expression:

$$\Delta x = e(y - y') \quad \dots (17.13)$$

The difference operator (Δ) shows the change in the value of a variable between two time periods. Notice that equation (17.13) can be derived from equation (17.2) by assuming that the public expects the previous year's inflation rate. That is, equation (17.13) can be viewed as an adaptive expectations version of equation (17.2).

Equations (17.11) and (17.12) suggest a positive relationship between inflation rate and output. When inflation rate increases, the Central Bank responds by increasing the real interest rate. This, in turn, contracts aggregate demand and total output in the economy. By using equation (17.12) in equation (17.11), we obtain the following equation of the aggregate demand (AD) curve:

$$y = y' + \left[\frac{c-dr'+dax'}{1+bd} \right] - \frac{dax}{1+bd} \quad \dots (17.14)$$

Aggregate demand is depicted in Fig. 17.2. Macroeconomic equilibrium will be attained at that point on the AD line where output and inflation rate are at their respective targets. Clearly, none of the variables will then have any tendency to change. Since output is pegged at its natural level, inflation rate will remain unchanged due to our formulation of the Phillips Curve in equation (17.12). Also, since output and inflation targets have been achieved, the Central Bank will have no incentive to change the rate of interest. Thus, the combination of y' and x' on the AD curve in Fig. 17.2 represents an equilibrium condition. We observe from equation (17.14) that if inflation is x' , output will be equal to y' when the following condition holds

$$r' = \frac{c}{d} \quad \dots (17.15)$$

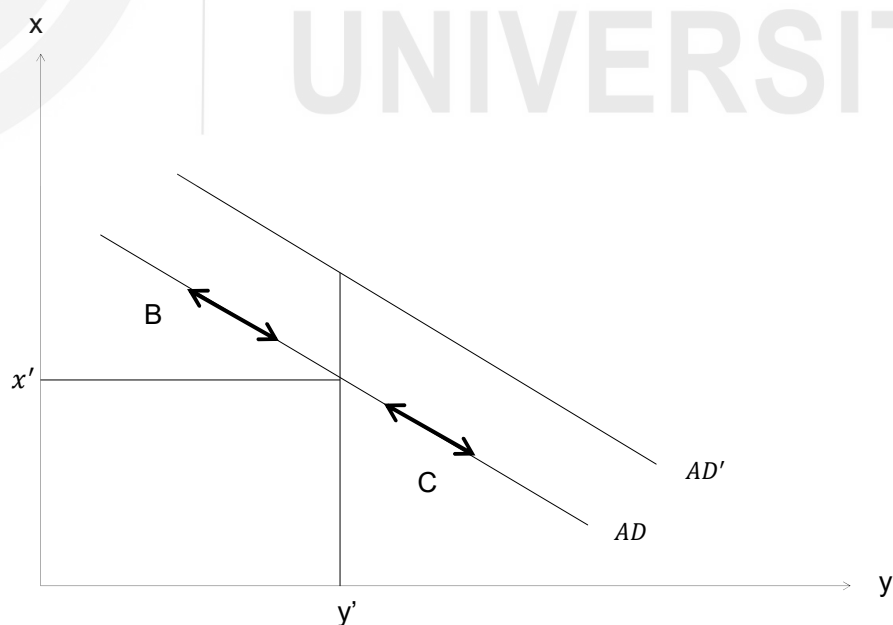


Fig. 17.2: Macroeconomic Equilibrium

The Central Bank must set parameter r' equal to the neutral rate of interest. This may not be simple, as the Central Bank may not be aware of the correct parameters of the IS curve and may end up underestimating or overestimating the neutral rate. Since $y = y'$ at the equilibrium point, it follows from equation (17.1) that:

$$r = r' = \frac{c}{d} \quad \dots (17.16)$$

If the Central Bank fixes the value of r' according to equation (17.15), the aggregate demand (AD) curve will pass through (y', x') in Fig. 17.2. If $r' \neq \frac{c}{d}$, then it must be the case that $x \neq x'$ when $y = y'$. This can be verified from equation (17.14). In Fig. 17.2, such a situation arises when aggregate demand is given by the line AD' .

Now, suppose the Central Bank fixes r' according to equation (17.15) so that aggregate demand is given by the AD line in Fig. 17.1. If the economy lies to the left of point (y', x') on the AD line, actual output will remain below its natural level. Therefore, due to our formulation of the Phillips' Curve, inflation rate must be falling. The Central Bank will then swing into action by reducing the rate of interest and, thereby, expanding aggregate demand. Thus, from B, the economy will move down along the AD line as indicated by the arrow pointed in the south-east direction. At B, actual output is less than the natural output and, thus, it follows from equation (17.12) that actual interest rate must exceed the natural interest rate fixed by equation (17.15). For any point to the right of (y', x') on the AD line, inflation must be rising causing our hypothetical economy to move in the north-west direction. Similarly, the actual interest rate must be less than the neutral interest rate. On the whole, the Central Bank has to first identify the correct neutral rate of interest and then vary the actual rate of interest towards the neutral rate.

Let us now examine the impact of two policy experiments: (i) an increase in inflation target of the Central Bank, and (ii) a shift in fiscal policy.

17.3.1 Impact of a Higher Inflation Target

Suppose the economy is initially at equilibrium point A in Fig. 17.3 when the Central Bank increases its inflation target x' . This causes the AD line to shift to AD' in Fig. 17.2. It can be verified from equation (17.5) that the intercept of AD line increases due to an increase in x' . Intuitively, this happens because a higher inflation target results in a lower rate of interest due to interest rate rule. Note that a shift in inflation target does not affect the neutral rate of interest in the economy, which remains specified according to equation (17.15). The first impact of a higher inflation target is felt by aggregate demand. There is an increase in aggregate demand as actual interest rate falls below its neutral level. This is captured by a movement from A to C in Fig. 17.3. At C, the actual output level exceeds the natural output level causing inflation rate to increase and output to fall along the AD' line. The new equilibrium is attained at B where inflation is at its new target, x'' . At B, actual output equals natural output and, since the